Predicting Airline Ticket Prices Using Statistical Learning

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I. Executive Summary

This project analyzes commercial flight pricing patterns using a large dataset containing over 300,000 flights collected from major Indian carriers in 2022. The primary objective was twofold: (1) identify the key drivers of flight ticket prices, and (2) develop a predictive model capable of estimating flight prices with high accuracy. Understanding these pricing patterns is valuable for consumers, airline analysts, and companies seeking to forecast travel costs or design pricing tools.

Exploratory analysis revealed that raw flight prices were highly skewed, with most economy flights priced between ₹3,000 - ₹10,000 (rupees) and business-class flights reaching above ₹100,000. Transforming the response variable, price, using the natural log provided a more stable distribution and improved the performance of statistical models. Boxplots and grouped summaries showed that three features—travel class, airline, and number of stops—were consistently associated with substantial differences in flight prices.

Multiple predictive modeling approaches were implemented, including Multiple Linear Regression, Elastic Net regularization, and a Random Forest tree-based model. Performance was evaluated using Root Mean Squared Error (RMSE) on both the log-scale and the original price scale. While the baseline linear regression and Elastic Net models performed similarly, the Random Forest model outperformed the others, achieving an RMSE of approximately ₹3,610 on the test set, which was less than half the error of the linear models. Variable importance results confirmed that class was the strongest predictor of ticket price, followed by airline, days left before departure, travel duration, and number of stops.

Based on overall accuracy and robustness, the Random Forest model was selected as the final model. It provides the most reliable price predictions and captures complex nonlinear relationships in the data, such as how price increases sharply with last-minute bookings.

This analysis demonstrates that machine learning can effectively predict flight prices and determine the factors that drive them. Future work could explore additional models such as gradient boosting, include other variables to expand the scope (ex: holidays or fuel prices), or test interaction variables.

II. Introduction

Airline ticket prices fluctuate widely due to changing pricing strategies, seasonal patterns, operational constraints, and consumer behavior. These fluctuations make it challenging for consumers to anticipate price changes and for businesses to understand the factors that drive pricing decisions. As travel demand continues to increase, the ability to analyze and forecast flight prices has become increasingly valuable in the aviation and travel industries.

This project focuses on examining the drivers of commercial flight prices and developing a predictive framework that can estimate price using commonly available flight attributes such as class, stops, duration, airline, and booking timing. Using a large dataset of domestic Indian flights from 2022, the analysis aims to uncover meaningful pricing patterns and construct reliable models that generalize to future flight listings.

The remainder of this report follows a structured analytical process, explaining the data characteristics, exploratory findings, model development, and evaluation results that lead to the final recommended model.

III. Business Problem

The goal of this project is to analyze the factors that influence commercial flight prices and build a predictive model that can estimate ticket prices using available flight attributes.

IV. Data Collection and Description

The dataset analyzed in this project was obtained from Kaggle named “Flight Price Prediction,” which contains commercial flight listings recorded in India during 2022. The full dataset includes 300,153 observations and 11 variables, each representing commonly available flight attributes such as airline, class type, number of stops, travel duration, and the number of days left before departure. The target variable is ticket price, expressed in Indian Rupees.

All variables were imported into R without missing values or duplicates. Categorical features, including airline, class, source city, destination city, departure time, arrival time, and number of stops, were stored as character variables and later converted to factors for modeling. Numerical features include duration (hours), days_left (days before departure), and price (₹). The dataset became well-structured and suitable for statistical learning after addressing the skewness of the price variable through a log transformation.

V. Exploratory Data Analysis

Exploratory analyses were conducted to understand the distribution of flight prices and identify key relationships with the response variable. The raw price distribution (figure 1) was highly right-skewed, with most fares concentrated at the lower end and a long tail of higher-priced flights. This skewness motivated the use of a log transformation to stabilize variance and better satisfy modeling assumptions.

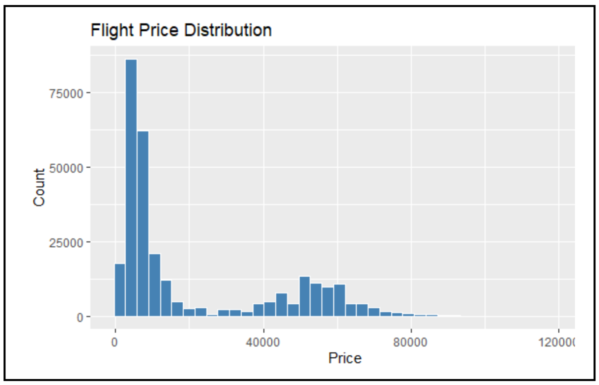

Fig. 1. Histogram of raw flight prices, showing a heavily right-skewed distribution

The log-transformed price variable exhibited a more symmetric distribution (figure 2), making it more usable for the modeling done in later sections. There are two peaks that can be interpreted as the mean for Economy class flights and the mean for Business class flights, since class is the main distinction in price.

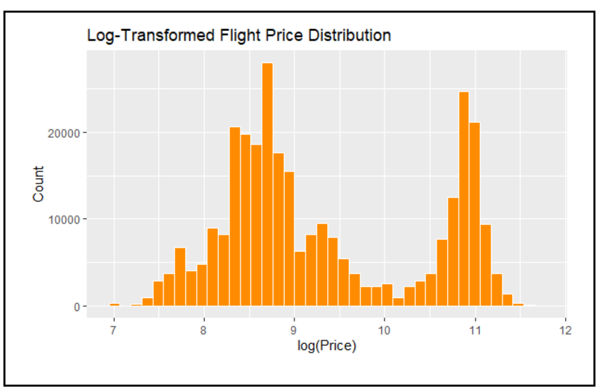

Fig. 2. Histogram of log-transformed flight prices, displaying a more symmetric distribution.

Since the modeling is performed on log(price), categorical variable comparisons (figure 3) were also evaluated on the transformed scale. Log(price) varied substantially by travel class, with business-class fares consistently higher than economy-class fares and showing greater variability. This indicates that class is likely to be one of the strongest predictors of price.

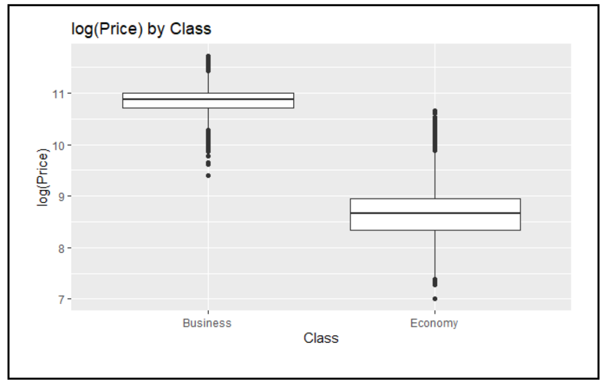

Fig. 3. Boxplot of log(price) by travel class, showing that business-class fares are substantially higher and more variable than economy fares.

Differences also emerged when comparing log(price) across the number of stops (figure 4). Non-stop flights tended to have lower log-prices, while flights with one stop displayed greater spread, and flights with two or more stops fell between the two in both median and variability. These patterns suggest that the number of stops contributes meaningfully to price differences.

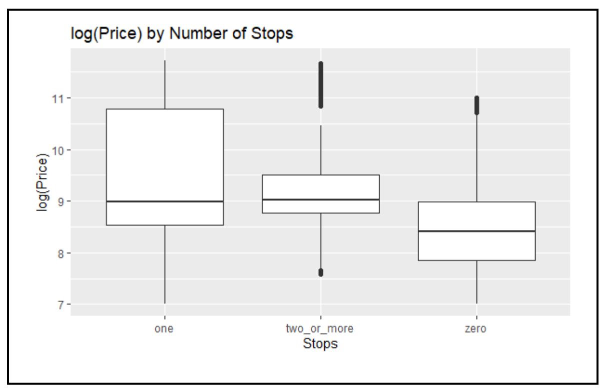

Fig. 4. Boxplot of log(price) by number of stops, illustrating that non-stop flights tend to be lower in price, while one-stop flights show greater spread and variability.

Overall, the exploratory results highlight several features—travel class, airline, number of stops, duration, and booking lead time—that appear strongly associated with ticket price. These insights guide the model construction and variable selection decisions in the next steps.

VI. Model Construction

Several statistical learning models were constructed to estimate flight prices using the log-transformed response variable. All categorical predictors were converted to factors, and reference levels were set to the most frequent categories within the training data to ensure interpretable coefficient estimates. To prevent inconsistencies between training and test sets, the same factor levels were applied across both subsets.

Three modeling approaches were implemented:

Multiple Linear Regression (Baseline Model)

A full multiple linear regression (MLR) model was fitted using all available predictors. This model serves as a baseline to assess linear relationships between log(price) and flight attributes while having the highest interpretability of the models for categorical and numeric variables.

Regularized Regression (Elastic Net)

To handle overlap among predictors and prevent unstable coefficient estimates, an Elastic Net regression model was fitted using the glmnet library. Elastic Net combines the strengths of both ridge and lasso regression, helping reduce overfitting and improve the model's ability to generalize to new data. Because glmnet is computationally intensive on very large datasets, a random sample of 25,000 observations was used for training. Cross-validation was then used to choose the best balance between the ridge and lasso penalties and to select the optimal tuning parameter.

Tree-Based Model (Random Forest)

A Random Forest model was implemented to capture nonlinear patterns and interactions that linear models may not fully represent. Using 200 trees and a limited number of predictors sampled at each split, the model was trained on the same 25,000 observation subset as the regularized models. Variable importance metrics were calculated to identify the most influential predictors of price.

VII. Model Evaluation and Comparison

Model performance was evaluated using both the log-transformed response and the original price scale to provide an interpretable assessment of prediction accuracy. Root Mean Squared Error (RMSE) and R^2 were used as the primary metrics. Table 1 summarizes the test-set performance for all three models implemented in this analysis:

Model <chr>	RMSE_log <dbl>	R2_log <dbl>	RMSE_price <dbl>
Baseline MLR	0.3240359	0.9152170	7776.630
Elastic Net (alpha = 0.5)	0.3242732	0.9150977	7703.144
Random Forest (ranger)	0.1847245	0.9724709	3609.997

Across the three modeling approaches, the Random Forest model achieved the strongest performance. It produced the lowest RMSE on both the log-scale and the original price scale, and the highest R^2 value (0.972), indicating that it explained more variation in log(price) than the linear models. Its RMSE of approximately ₹3,610 represents less than half the prediction error of the other two models, demonstrating a significant improvement in predictive accuracy.

The baseline Multiple Linear Regression model performed reasonably well, with R^2 around 0.915, but its higher error suggests that linear relationships alone are not sufficient to capture the complexity of flight pricing. The Elastic Net model yielded performance similar to the MLR model, showing that regularization helped stabilize the coefficients but did not improve predictive accuracy by much.

Overall, the results indicate that nonlinear, tree-based methods are better suited for this dataset, likely because they can capture interactions and complex relationships among flight variables that linear models cannot fully represent.

VIII. Final Model Selection

Based on the test-set evaluation results, the Random Forest model was selected as the final model for predicting flight prices. Among all algorithms tested, it achieved the lowest RMSE on both the log scale and the original price scale, as well as the highest R^2 value, indicating that it captured more variability in flight prices than the linear and regularized regression models.

The Random Forest model is well-suited to this dataset because it can automatically account for nonlinear relationships and interactions among flight features, such as how price varies jointly with class, airline, booking timing, and duration. These complex patterns are difficult for linear models to represent effectively, which explains the performance gap in the comparison table. The Random Forest's variable importance results highlighted class, airline, days left before departure, and flight duration as the strongest predictors, which matches the trends identified during the exploratory analysis.

Although Random Forest models are less interpretable than linear models, their superior predictive accuracy makes them the most reliable choice for estimating ticket prices in this context.

IX. Conclusion

This project analyzed the factors influencing commercial flight prices and developed predictive models capable of estimating fares using commonly available flight attributes. After evaluating multiple statistical learning methods, the Random Forest model delivered the strongest performance, achieving the lowest prediction error and highest explanatory power on the test set. Its ability to capture nonlinear relationships and interactions allowed it to outperform both the baseline multiple linear regression and the Elastic Net model. The most influential predictors—such as travel class, airline, days left before departure, and flight duration—aligned closely with the trends identified during the exploratory analysis.